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Reading with Robots: A Research-through-Design Study

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Abstract
In recent years, there has been a significant decline in reading motivation among Swedish school children. This research-through-design study explored if and how talking to a social robot about books might help to motivate fourth-grade students (10–11 years old) to read more. The study followed a three-phase design process, in which we designed and tested robot book talks in a school class with 19 students, using a Wizard of Oz set-up with a Furhat robot. In the book talks, the robot's level of engagement varied between passive and active. In the passive mode, which functioned as a baseline, the robot asked general questions but made relatively little effort to make sure the students were concentrating on the task. In the active mode, on the other hand, based on design ideas developed throughout the design process, the robot asked questions directly related to the students' books and guided them through their task. The results indicate that the interaction with the robot was appreciated by most students and that robot book talks have potential to contribute to fulfilling basic psychological needs of competence and relatedness, which according to self-determination theory are fundamental to motivation and well-being.

Keywords Reading · Motivation · Interaction design · Social robots · Self-determination theory · Education

1 Introduction

This paper reports on an exploratory interaction design study responding to a growing disinterest in reading among Swedish school children. As of 2021, only 18% of 11-year-olds in Sweden stated that they "like reading a lot". This is a relatively small proportion within this age-group in in comparison to the other EU/OECD countries [1]. Moreover, only a minority of children in Swedish upper middle school and lower secondary school feel motivated to read on their own [2], and reading in those grades has decreased since 2007 [3].

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The study presented in this paper explored if and how talking to a social robot about books might help to motivate students in school year 4 (10–11 years old) to read more. The study followed a Research-through-Design (RtD) approach, which means that a design project formed the backbone of the research. From a design perspective, it is central to articulate target user experiences of the reading robot that are appropriate for a middle-school classroom situation. Such target experiences are referred to as user experience (UX) goals, interaction design objectives, or desirable use qualities [4–6]. Establishing UX goals and evaluating the experience is often overlooked in human-robot interaction (HRI) research [7], but constitutes the starting point for the work presented here.

The overarching question for the study therefore is if and how social robots might be used to motivate children within this age group to read. This can be broken down into three research questions, each one corresponding to a phase in the design project that forms the backbone of the study:

- **RQ1:** What are the UX goals for a social robot intended to motivate children to read?
- **RQ2:** How should a social robot be designed to motivate children to read more?

- **RQ3:** Does the proposed design meet the UX goals?

The remainder of this paper is structured as follows: Section 2 provides some background on (a) self-determination theory and its view of human motivation, on (b) reading motivation in school children, and (c) the use of social robots in educational settings. Section 3 outlines the Research-through-Design research approach and methodology that underlies this work. Section 4 describes in detail the design process, consisting of Exploration, Generation and Evaluation stages, as well as the results of our study. Section 5, finally, presents some discussion and conclusions.

2 Background

The following three subsections set the scene by providing some background on the general view of motivation that is underlying this research, how this applies to reading motivation, and how social robots are being used in educational settings.

2.1 Self-Determination Theory

Positive UX is related to the fulfilment of users' psychological needs such as relatedness, stimulation, and competence [8]. Self-determination theory (SDT) [9] is a wide-ranging psychological theory addressing human motivation, personality, and well-being. Applying central SDT concepts to reading, you could say that someone who reads autonomously because they find it enjoyable is *intrinsically motivated* to do so [2, 10, 11]. A person who is *extrinsically motivated* to read, on the other hand, does so because they believe that it will help them achieve an external goal [12], such as a good grade.

The disadvantage of extrinsic motivation is that rewards and punishments can be controlling, which then can lead to lower performance [10, 13]. There are, however, positive forms of extrinsic motivation that do indeed spur autonomous behaviour. An example would be if somebody sees value in reading, or identifies themselves as being a reader, without necessarily being interested in it or enjoying it [9, 13]. Importantly, SDT identifies three essential psychological needs for well-being that are also central for motivation [13]:

- *autonomy:* a sense of initiative and ownership in one's actions
- *competence:* a feeling of mastery and a sense that one can succeed and grow
- *relatedness*: a sense of belonging and connection

SDT favours structure over control. In the educational context, structure concerns how well teachers organize an environment that is challenging enough, but also provides guidance that satisfies the students without controlling them. Hence, a well-structured learning environment is characterized by the following three aspects [14]:

- clear goals, rules, and expectations *before* a learning activity
- help, guidance, and supervision *during* a learning activity
- positive, constructive feedback *after* a learning activity

2.2 Reading Motivation

Research has shown that students who are intrinsically motivated often do better in school [2, 10–12]. Intrinsic motivation is closely tied to reading competence [15], and children who do not develop intrinsic reading motivation usually remain poor, slow, or struggling readers [16].

Supporting relatedness in school may be a key solution to promoting intrinsic motivation, as the social learning environment plays a large role in education [17]. Conversations about books, culture, politics, or homework with peers or role models have also been shown to facilitate reading motivation in children [12, 18]. One way to achieve a positive learning environment while providing structure, is to let students choose what books to read [2, 15, 18]. Another important aspect for students' reading habits, abilities, and engagement is how teachers work to inspire and sustain motivation [19]. This makes sense from an SDT standpoint, although providing autonomy-supportive teaching is not always easy, due to time constraints or lack of resources, and for the simple reason that teachers also have basic psychological needs [13]. This challenge provides an opportunity for social robots in education.

2.3 Social Robots in Education

Social robots in educational settings have been successful in creating affective and cognitive outcomes, which indicates that robots may indeed play a role in the modern classroom [20, 21]. Children can be engaged by social robots as long as the reading activities are both challenging and diverse enough [22]. Some studies have begun to examine the use of robots to support children's reading [23–25]. Michaelis and Mutlu [24], for example, found that early adolescent children (10–12 years) enjoyed the presence of a learning-companion robot while reading and reported that the robot made reading "less boring". Caruana et al. [25] found that children evaluated the robots in their study as being intelligent enough to read, listen and comprehend stories and

concluded that reading robots can provide an engaging and non-judgmental social context to promote reading engagement. These findings indicate that robots have potential as reading partners for children. One of the main contributions of our study is that it goes goes beyond the existing literature by establishing and evaluating relevant UX goals for reading robots through a rigorous and well-established design process, as described in detail in the following sections.

3 Research Approach and Methodology

As this is a Research-through-Design study, the backbone of the method is a design process characterized by generative ideation, concept development, and making [26]. The aim of this kind of research is to produce and iteratively refine designed products that also advance the development of knowledge, in this case knowledge on how to make use of social robots in the context of motivating to school children to read more. The process of designing forces the design researchers to confront tensions between conflicting knowledge, theory and technology, ideals and reality. When carefully documented and analysed, the RtD process can articulate tacit values and latent needs, as well as patterns for and features of solutions that are embodied in the designed artifacts [26]. These can then be used as design examplars for future design work by others. It might be worth noting that applicability of RtD findings is not necessarily intended to be general. Instead, transfer of knowledge needs to be carried out with a degree of judgement on a case-to-case basis [26]. RtD is explorative or 'experimental' in the broader sense of trying things out to see if they work, but not in natural science sense of a lab study with controlled and isolated variables, where a hypothesis derived from theory is corroborated or rejected [26]. This means, RtD is a research approach that can be characterised as building on an action-oriented, reflective, and mainly qualitative methodology, often within a pragmatist tradition.



Fig. 1 The Furhat Robot (http://www.furhatrobotics.com). Copyright 2023 by Furhat Robotics. Reprinted with permission

The design process in the study documented here followed the first three stages of Cross' basic four-stage model of the design process [27]: *exploration* of an ill-defined problem space; *generation* of concepts; and *evaluation* against goals, constraints, and criteria. There is, in Cross' model, an iteration between the evaluation stage and the generation stage, since the generated concepts are not always satisfactory. The fourth stage in Cross' model is *communication*, and writing this paper is part of that. In this study, the object for design and research is the activity of the children's book talks with the robot.

The social robot used for the present study is Furhat from Furhat Robotics [21, 28] (see Fig. 1). The use of back-projection onto a firm, translucent mask gives Furhat an animated face [28], the ability to make facial gestures [29], and thus to perform some form of non-verbal communication, which is vital for social agents [30]. Inside Furhat are a processor, a speaker, a camera, a microphone, and motors that allow it to look around [31]. The speaker enables Furhat to use synthesized speech [32]. As Furhat is 41 centimetres tall, the head is roughly human-size and can stand steadily on a table, it can be placed at eye-sight level for middle school students, which makes it easy for them to perceive it a friendly peer [33].

During the book talks with students, the robot was controlled using the Wizard of Oz method, which means that the behaviour and responses of the robot were simulated and controlled by a human operator without the participants' knowing beforehand [34]. Notes and recordings of book talks and interviews were transcribed and analysed using ethnographic and latent thematic analysis [35, 36].

Everyone who took part in this study participated voluntarily. All participating adults and students (and their legal guardians) were informed about their respective roles in the study. Together with their legal guardians, the students signed an informed-consent form handed to them via their teacher.

4 Design Process and Results

The procedures and results of the three stages are described in detail in the following subsections. The overall process was structured as follows:

1. Exploration
   (a) Literature analysis
   (b) Site visits
   (c) Definition of UX goals

2. Generation

   (a) Concept development
   (b) Expert review
   (c) Character development

3. Evaluation

   (a) Field test of Wizard of Oz prototypes
   (b) Thematic analysis of test results

4.1 The Exploration Stage

The exploration stage involved both an analysis of existing literature on related work as well as site visits to the collaborating school to develop UX goals, addressing the first research question: *What are the user experience goals for a social robot intended to motivate children to read?*

4.1.1 Participants

The participants included a teacher (female, age 47) and 19 students in school year 4 (age 10–11 years, 7 female and 12 male). The school had 150–200 students from pre-school to school year 6. It was located at the border between a middle-class area with mostly standalone houses and residents of Swedish decent, and a working-class area with mostly apartments and a large portion of residents with an immigrant background. Little more than half of the students reached the learning objectives in year 6, which was low compared to other schools in the municipality.

The students' reading comprehension level was close to average ($M = 5.32$, $SD = 1.42$, range: 2–8), as measured using the DLS 4–6 test, which is a Swedish diagnostic tool for analyzing reading and writing abilities in school years 4–6. It was used by the teacher to test the students' reading comprehension levels as part of the regular teaching activies. Based on prior norm data, the test results of the DLS are scaled using the Stanine technique on a nine-point standard scale with a mean of five and a standard deviation of two.

A background questionnaire about the students attitudes' towards reading showed that they liked silent reading in a book ($M = 4.26$ on a 5-point Likert scale, $SD = 1.19$, $n = 17$). 12 out of 19 students stated that reading was fun, and 14 out of 19 characterized it as important or necessary, while 4 out of 19 described it as boring. 15 of 19 students stated that they read "often" or "a lot".

4.1.2 Procedure

The UX goals were initially based on the literature on reading education, reading motivation, self-determination theory, and social robots (in educational settings), as described earlier. The first author also carried out a site visit to the school class that would participate in the study, and the second author had visited the same class five times prior to this study. During the site visit, the first author took notes regarding the children's behaviour and how their teachers communicated with them. Notes on the readings and notes from the site visit were jointly analysed using thematic analysis [35, 36], which formed the basis for the four UX goals listed below. The readings had greater impact on the development of the UX goals than the site visit, but the site visit allowed the designer (the first author) to familiarize himself with the students, and vice versa. This constitutes an important part of building rapport and a feeling of safety for the upcoming evaluations and for building a general understanding of the classroom context. The goals were also adjusted throughout the design process, although the core ideas remained the same.

4.1.3 Exploration Results

Four UX goals, described below, were defined. An assumption underlying all UX goals at this stage was that the students have the option to choose books that they want to read, to fulfil the need for autonomy, which is a key component according to SDT [13]. The goals of the interaction with the robot also had to be clearly stated for the students in order to create a highly structured setting [14].

- **UX Goal 1:** The agent[1] should seem effective enough at communicating to maintain *adaptive* conversations.

The flow of conversation is important, as a fluid conversation with the robot may encourage children to reflect on the contents of the book and discuss it in a collaborative manner. Unpredictable and inconsistent responses can lead to an experience of diminishing autonomy [37]. Moreover, effective agent instruction also requires non-verbal communication [30]. Adaptive feedback in conversations about books may help the children to find relatedness, display and feel their own competence, and meaningful interest in reading and the discussion of literature, which should improve motivation in reading [12, 18]. This is also supported by SDT [13].

- **UX Goal 2:** The agent should appear *trustworthy* enough to be viewed as a peer that the children can explain things to.

[1] The term 'agent' is used instead of social robot since the decision to design a physical robot had not yet been made at this stage.

The children need to trust the agent to be encouraged to teach it things. For this to be possible, the agent should be non-judgemental and avoid correcting the student. If the agent is presented as less knowledgeable than the student, the student will be able to *learn by teaching*, which can help to boost the confidence of weaker readers [33]. Employing politeness and behaviours that build rapport are known to positively impact users [17].

- **UX Goal 3:** The agent should be able to create a sense of *relatedness* among the students.

Book talks with the agent should engage the students enough to make them feel connected to one another, both in that they are teaching the agent together, but also in their competence. This way they may also learn from each other. During the site visit, some students were quite curious about the robot and asked questions such as: "Is the robot human-like?", "Can it write?", "Can it talk?", "What colour is the robot?", "How tall is it?". This highlighted the risk as well as the potential of the "novelty effect", which means that having a robot in the classroom might have a positive impact just because it is something new and different [28, 38, 39]. Ideally, the effect would make the students more interested in working with the agent, however it could also cause them to be more curious about the robot as such than the tasks they are supposed to perform in interaction with it. The idea behind UX Goal 3 is that relatedness (among the students) would make the book-talk more stimulating than the novel sensation of talking to a robot.

- **UX Goal 4:** The agent should be able to change its appearance and demeanour to better fit the book, thus making the book talks *immersive*.

To immerse or invite the students into "the world of the book", the agent should be able to change its characteristics according to the book. This is intended to make the book talks more engaging and should reinforce the students' experience of competence and relatedness, which are two of the key components of SDT [13]. This UX goal is also supported by previous literature on the importance of tailoring the visual and auditory appearance of agents to specific learning contexts to promote student motivation and improve learning outcomes in educational settings [40]. Children read books partly to become immersed in the story [19], and having an agent that could change in accordance with the book might contribute to immersion.

4.2 The Generation Stage

The generation stage consisted of concept development, an expert review, and character development. It addressed the second research question: *How should a social robot be designed to motivate children to read?*

4.2.1 Participants

The first and second author participated in the development of concepts. Furthermore, there were also three participants in the expert review: two researchers in educational practice (female, 45 and 59 years old) and the teacher who also participated in the site visit during the exploration phase (female, 47 years old).

4.2.2 Materials

About 30 different ideas were initially noted down for consideration. As these ideas were developed alongside the UX goals, they decreased in number as the goals were gradually clarified.

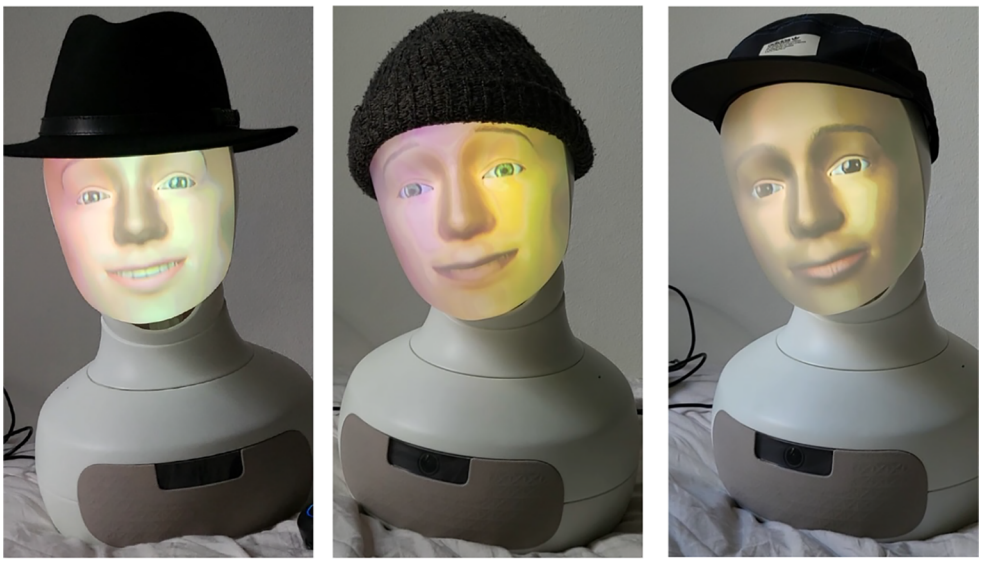
Out of the initial 30 ideas, 16 design concepts were developed. They included robots/agents in roles such as tutor, peer, mediator, and student. Concepts that did not allow for group interaction with the robot or were not possible to prototype using Wizard of Oz were eventually discarded or combined to create new concepts. Four concepts remained to be reviewed with experts: (1) *The Quiz Agent*—a tutoring robot that would quiz students on the book they had read; making reading more interesting; (2) *The Agent that Changes Shape*—shape-shifting could be used to immerse students in the world of the book; (3) *The Encouraging Little Robot*—a robot with which students could discuss what they have read as the robot could ask about their experience and recommends ways to make reading more enjoyable; and (4) *Learning through Teaching with a Robot Student*—this robot has read the same text as the students, but has problems understanding things such as idioms and metaphors, allowing the students to teach it.

These four concepts were then subjected to an expert review (as described in more detail in the following subsection), leading to a final concept, which was to be tested in the evaluation stage.

4.2.3 Procedure

The expert review was built around the PMI method [41]. PMI stands for *plus, minus*, and *interesting*, and the aim of the method is to review concepts in terms of advantages, disadvantages, and interesting aspects. It was held as a video conference due to pandemic restrictions at the time. After

Fig. 2 The three Furhat characters



(a) Filip (b) Freja (c) Ted



Fig. 3 A group of students interacting with the Furhat robot (the photo is traced for anonymization)

the PMI session, the first and second authors and the participants discussed how the concepts could be implemented. Video and audio were recorded, and notes were taken. After the workshop, the concepts were synthesised based on the participants' contributions, eventually resulting in the final concept.

4.2.4 Generation Results

The reviewers were positive that the Shapeshifting and Learning through Teaching concepts could be combined. This resulted in the choice of the Furhat robot as the implementation platform and the creation of the final concept: *The Robot Book Talk*. The reviewers argued that the robot should *never correct the students but rather guide them through questions*. This was added to UX goal 2.

The robot was to take the role of a peer that would learn alongside the students, allowing the students to feel competent, which is in line with SDT. The students would have to collaborate during book talks to explain things to the robot, since the robot did not know everything about human culture and language. This was, with reference to SDT, supposed to contribute to relatedness between the students. Autonomy, the third psychological need in SDT, was in this concept catered for by allowing the children to choose which book to read.

Presented below are the three Furhat characters, whose faces and voices were chosen to match the ages of the students (see Fig. 2). The books to be read were chosen by the teacher of the school class that would participate in the evaluation stage. The three books inspired the three characters that were developed.

"Black Belt" with Filip Black Belt (Swedish title "Svart bälte") by Kerstin Lundberg Hahn is an action-suspense book about a judo apprentice and his friends, who go out to find money that was stolen from the judo club. The character Filip wants to look tough, but is friendly. He likes to read suspenseful books and is laid-back. He enjoys talking about books with his friends.

"Semlan and Gordon" with Freja Semlan and Gordon: The Dad with the Big Shoes (Swedish title: "Semlan och Gordon: Pappan med de stora skorna") by Moni Nilsson is a dramatic but humoristic book about Semlan, a girl caught in the middle during her parents' divorce. The character Freja is relaxed but contemplative. She likes to read and talk about funny books. She often asks questions that allow her friends to speculate more.

"Go Out with Me" with Ted Go Out with Me (Swedish title: "Bli ihop") by Thomas Halling is a realistic and lighthearted book about Kalle, who falls in love for the first time

but does not know how to initiate a relationship. The character Ted is happy and curious, albeit a bit nervous. He enjoys reading as it helps him learn about the world.

4.3 The Evaluation Stage

The evaluation stage consisted of a two-session qualitative field test at a school that addressed the third research question: *Does the proposed design meet the UX goals?* The two sessions were carried with about a week between them.

4.3.1 Participants

The same 19 students (12 male and 7 female, age 10–11) as in the site visit during the exploration stage participated in the evaluation of the experience of having book talks with a social robot. 15 students were present for the introduction. The teacher divided the class into groups. Groups 1–5 consisted of three children each. The four students who missed the introduction were later (the following week) put into an additional Group 6, participating in only one session. The students will in the following be referred to by their group number and a letter, e.g., 1A, 1B and 1C representing the three individual members of Group 1.

4.3.2 Materials

The wizard used the software development kit (SDK) from Furhat to control the robot. It allowed control of facial expressions and movement as well as utterances. Predefined phrases were cut-and-pasted from an Excel sheet into the text-to-speech interface. The wizard also typed improvised responses that were not pre-defined. The wizard practiced controlling the robot in two pilot tests and changed pre-defined phrases so that the speech synthesis would pronounce things correctly and have a fluidity of conversation.

The design was tested in two modes or variations: *passive* and *active* engagement in the book talk. The rationale for this was that a passive mode that would be much easier to build and implement, but we were not sure if it would work as good as a robot that engaged more in book talk. The active mode would be a teachable agent that made the book talk more structured, which would be favourable according to SDT.

In the *passive mode*, the robot asked general questions, but made relatively little effort to make sure the students were concentrating on the task. Moreover, the character introduction was shorter, as the robot would not ask the students to help them if they misunderstood anything in the text. In this mode, the robot was less engaged in discussing the books and asked general questions revolving around the students' opinions on the texts. It could still ask follow-up questions but would not guide the students to help them focus. The passive mode served as a baseline that the designed active mode could be compared to. The *active mode*, on the other hand, built on the ideas developed throughout the design process including the idea of the robot as a teachable agent. Here, the robot asked questions directly related to the contents of the students' books and made more effort to guide the students. The interview guide for the semi-structured interview sessions covered the following areas: description/introduction of the robot, motivation for choosing the book and character, impressions of the book talk, what the robot did and how it contributed to the discussion; motivation for amount to read, preference for meeting the same character again, and perceived learning.

4.3.3 Procedure

The evaluation stage started one day before the first book talk session with the teacher introducing all three books and showing the students prerecorded videos of the Furhat characters introducing themselves. Since there were only a few copies available of each book, and many of the groups wanted to read Black Belt, several groups were assigned their book through a lottery. The students read their books individually (without interacting with the robot or other students).

All groups except Group 6 met Furhat in both active and passive mode. In the first session, Groups 2 and 4 met Furhat in passive mode and the remaining groups (1, 3, 5) met Furhat in active mode. The following week the modes were reversed for the second session (Group 6 only met Furhat once, in active mode). Both sessions, regardless of mode, started with the first author welcoming the students to the session and explaining the procedure. To hide the fact that the first author was controlling Furhat's movements and dialogue, the students were informed that he was taking notes on the interaction. The subsequent structure of the sessions varied depending on what mode Furhat was in.

In the *active mode* sessions, the Furhat characters introduced themselves and stated that they had read the same books as the students had, but since they knew little about human culture and language, they had not understood all of it. Furhat would therefore ask the students to summarise what they had read and then explain some events in the book (e.g., "Why did Semlan get angry with his parents?"). If the students lost focus while answering the questions, Furhat would repeat a question or ask them to elaborate on something that they had previously said. These questions were followed by a general discussion, where Furhat asked for the students' thoughts and reflections on the book. In the active mode, Furhat also initiated the discussion by asking questions that were tailored to each book.

In the *passive mode* sessions, the Furhat characters would not ask the students to help them out if they misunderstood or wanted help with interpreting the texts. Moreover, the discussion questions were more general (e.g., "What did you like most about the book?"). As a consequence, the overall contribution of Furhat to the book talk was much smaller in the passive mode compared to the active mode.

As the Wizard of Oz method enabled Furhat to react to what the students were saying, the number of questions asked in both active and passive sessions varied between individual sessions. Each book talk session lasted a maximum of ten minutes. After this, the students were given two minutes to decide how much they would read for their next book talk the following week. The only requirement was that they had to read at least as much as they had read for their first book talk. Once they had decided, the semi-structured group interview was conducted using the questions described in the previous subsection. After all groups had participated in the second book talk, the first author debriefed the students about the study. The debriefing included a demonstration of the Furhat software development kit, how to change Furhat's facial expression and voices, and how the Wizard of Oz method worked, following the recommendations of Rea et al. [42] and Rosén et al. [43].

In line with ethnographic thematic analysis [36], video recordings from both the book talks and the interviews, were transcribed using a selective transcription technique. This meant that episodes with higher thematic relevance were transcribed in detail, whereas episodes with less relevance were briefly summarised. Transcriptions were then analyzed in relation to the research questions as well as theory. Initially identified codes were: Friendship with Furhat; Complications; Competence; Connectedness; Reading Enjoyment; Immersion; Not Used to Robots; Autonomy; Good or Interesting Things About Book Talks; Curiosity and Nervousness; and finally Characterisation, Who's Writing? and Technology Interest. These initial codes were later refined into the final themes described below in the Evaluation Results.

4.3.4 Evaluation Results

The results of the thematic analysis of the qualitative data from the book talks and interviews are described in the following. The final themes were Relationship with the Robot; Autonomy and Enjoyment; Competence; Relatedness; Immersion; Characterisation; Perceived Learning; and Task Engagement.

Relationship with the Robot Despite initially being nervous around the robot, all groups were comfortable after some time of interaction. 1C said that they were initially made uneasy by the fact that Furhat appeared a bit incoherent and moved fast. The groups' enjoyment of interacting with Furhat significantly increased over time, though. For example, groups 1, 3 and 4 explicitly stated that their second meeting with the robot was more enjoyable, because they knew the respective characters already. 5C may also have become attached (perhaps playfully) to the robot, as they said "I'm sorry, Ted, we will never meet again" to the robot toward the end of their second talk, acknowledging its characterisation as Ted. There were no notable differences between the active and passive mode in the relationship formed with the robot.

Autonomy and Enjoyment Most groups chose to read at least one more chapter for their second book talk, than they had read for the first talk. Group 6 was an exception. They said that they would have read the same number of chapters for their next book talk, had there been another session. Group 5 finished their books before the second session by reading four chapters instead of the three chapters they had originally opted for. While most students found the reading enjoyable, Group 1 was the only group where all members were very pleased with their book, "Black Belt." Despite this, 1C said they did not necessarily think that book talks had any effect on their reading motivation, as they likely would have read it either way. 5C thought that "Go Out with Me" had too little text in it. Further, while most of the students were alright with the book they had been assigned in the lottery, 2A and 2B were disappointed. Their opinions on "Semlan and Gordon" were reflected in their relative lack of interest in Furhat, and likely their motivation to read as well. "Why are we even doing this?", 2B asked at one point. These opinions show that even though students may in fact enjoy reading, reading motivation may require that they are able to choose a book of their own liking. The autonomy of the students was affected by the fact that there were only few copies of "Black Belt," which many students wanted to read. For 2A and 2B, this also affected reading enjoyment. There were no notable differences regarding autonomy and reading enjoyment or motivation between the active and the passive mode.

Competence Overall, book talks flowed better in active mode, as Furhat's questions guided the students through their task. In passive mode, Furhat would never initiate discussion around any specific plot element and would instead leave that responsibility to the students. For example, Group 4 was noticeably less engaged in their passive mode session where Freja asked them to explain why they thought their book was funny. At first, instead of answering, they were looking around in the room. When Freja once again asked "Why?" 4B responded: "I don't know. It was good.

It was about a lot of things. Both outdoors and indoors, and when [Semlan and Gordon] were little and things like that." During their active mode book talk, Group 4 were quick to give nuanced answers, e.g., when Freja asked what it meant to betray someone's trust, they explained that it meant that you have promised somebody something, without living up to your promises. In active mode, Freja guided the students while still allowing them to explain things to her that she did not understand. The interaction indicates accordingly that they perceived her as a peer even though she guided the interaction.

Relatedness Book talks in active mode were clearly better in terms of task-relevant relatedness. In comparison, the relatedness in passive mode was mostly observed in moments when students were distracted from the task. In active mode, Groups 1, 3 and 4 showed relatedness in that they took turns answering Furhat's questions and added to each other's answers. While they may have given Furhat the wrong definition of some expressions that were asked about, their facial gestures and body language indicated that they enjoyed collaborating to answer the questions. To further support relatedness, and because it was designed to be non-judgemental, Furhat never corrected them. These task-relevant moments of relatedness between students seem to have had positive effects on the entire book talks, as both Group 1 and 3 were eager to find all the other passages that Filip was referring to in his questions. Once one of the group members had found the correct passage, they showed it to their peers. Another indication of relatedness was when Group 4 smiled at each other while talking about what they had read together.

Immersion Indicators of immersion and investment were more apparent in the active mode. The Furhat characters asked the students to talk about specific scenes in the books, which allowed them to take the book characters' perspective. For example, Group 1 and 3 were looking forward to finding out how the story would progress, because they were immersed in the world of the book. When Filip made a facial gesture indicating fear, to show how he believed a character in the book "Black Belt" looked when she spotted the mother with the stroller, 3A and 3C started discussing how they thought the mother felt at that very moment.

Characterisation The characterisation of the Furhat characters seemed to be of importance, as the students were curious to learn more about the characters. Moreover, Freja being a girl was a reason why 2C chose the book "Semlan and Gordon." This implies that characteristics such as gender can affect how a student chooses to engage with them, if they have the autonomy to choose. Conversely, 2B was not happy with "Semlan and Gordon," and neutral to the character of Freja. 2B wanted to "be around […] robots" but disregarded their characterisation. Comparing these two comments it becomes evident that the students applied some suspension of disbelief to the characteristics of the Furhat characters. For example, Group 3 talked about Filip's feelings and asked if he had been to the Liseberg amusement park in Gothenburg, and 5C compared Furhat to the literary character Alfie Atkins (Swe. Alfons Å berg) because he behaved like a normal Swedish boy, i.e., "a Svensson". At the same time as acknowledging that Furhat behaved almost like a real person, 4B recognised that the behaviour of the robot was a product of mechanics and coding. On their first interview, 4B commented that Furhat's programming was good, as it "could do many things, and talked well", "It can blink, it's like a face. It can ask questions", but also, "it was a little funny too. It talked funny. And it was nice. Like kind. It was pleasant." Regardless of mode, some students were very interested in Furhat itself, and whether the first author was controlling it (which he was). After Group 3 found out that Filip really liked pancakes, 3B asked the first author if their favourite food was pancakes. 3B tried this strategy again when they found out that Filip's favourite colour was red. After the book talk was finished they tried to reveal the wizard by pointing at the first author and saying: "You typed what he was going to say!" At another point, 3A tried to find out whether Filip could read by putting their copy of the book in front of the robot's camera. 2A and 2B also tried to reveal the wizard, saying that they knew that the first author was controlling Furhat. 5C and 6B asked the author if this was the case but accepted when the first author declined to comment on the matter. Despite their suspicion, 5C and 6B would both still talk to Ted as though he was a real person for the remainder of the book talk.

Perceived Learning Some of the students stated that they had enjoyed the book talks, both for the reading and for the interaction with the robot. 2C and 4A found that the book talks had been useful because discussing it afterwards made it easier for them to remember what they had read. Most of the participating students either answered that they had learned things about robots or would not specify what they had learned.

Task Engagement The students were not always concentrating on the task. Group 6 showed signs of disinterest in the book talks. Aside from a few comments from the other group members, 6B was the only one of the students in Group 6 who spoke throughout the talk. It seemed that the group was more interested in the robot itself, and they reacted positively to Ted's grimaces. In both of their sessions, Group 2 had problems focusing on the task. Their active session was

slightly better, as Freja guided them with her questions, but even then, their answers became gradually less serious. 2A also had an impolite attitude toward Furhat, despite being genuinely interested in the robot. When they started discussing a scene in "Semlan and Gordon" where Semlan spat at a woman, 2A looked at Furhat and said "I could spit at you" in a mocking tone. A potential factor in the lack engagement from 2A and 2B may have been a lack of autonomy in choosing what book to read for themselves. Some groups seemed to need more guidance than others. In this respect, the active mode is likely to be better than the passive one.

5 Discussion and Conclusions

This section discusses findings and limitations of our study, presents ideas for future work, and draws some conclusions.

5.1 Findings

All students wanted to speak to Furhat again and to experience all the characters, which indicates that the book talks established at least some extrinsic motivation [2, 10–12] to read, regardless of mode. The book talks might also have allowed the students to fulfil a need for relatedness which is a motivational factor [9].

A couple of students were visibly disappointed with not having the autonomy to choose "Black Belt" for their book talks. Reduced autonomy can result in people doing things solely because they must [9, 10]. Comments such as "Why are we even doing this?" suggest a controlling type of extrinsic motivation [2]. To fulfil all students' need for autonomy, a sufficient number of copies of each book is necessary to cater for autonomy and hence intrinsic motivation [15, 19]. With choice of reading, robot book talks can combine structure [14] and reading autonomy. Also related to autonomy, Furhat's fast movements and incoherent speech made one student nervous upon first meeting the robot. Complications in agent-user conversation can lead to unease and reduced autonomy [37].

The active mode was more effective than the passive one in letting the students show their competence. In active mode, the students discussed their own interpretation of the books and gave nuanced answers to Furhat's questions. This shows that there were moments that suggest that an active robot book talk can support competence. As reading is strongly tied to self-concept [2, 15, 44], a method that enables students to show their competence would be beneficial.

The active mode also provided stronger support of relatedness than the passive mode. In line with the ideas suggested by Vansteenkiste et al. [14], that is, providing high-structure through clear goals, feedback, and guidance, the questions in active mode created opportunities for the students to connect when performing their task. The introduction provided by the teacher was also important in setting expectations, goals and rules. What was perhaps lacking in relation to the high-structure principles put fourth by Vansteenkiste et al. was constructive feedback after the book talks. In terms of UX, the high-structure principles of goals, guidance, and feedback contribute to a feeling of safety without control. This means that high-structure provides the limits of leeway within which the students can have autonomy, feel competent, and by working on the same thing also connect to each other and experience a sense of relatedness. A guiding and active interaction with the robot can contribute to this.

The moments of relatedness seen in the results are promising, as they imply that robot book talks can provide opportunities for students to offer each other support and inspiration to higher achievement in school [12, 18], and to create a learning environment like the one described by Walker and Ogan [17]. Over time, robot book talks where relatedness is strong could be used to challenge students, by having them read and discuss books that they would not have read for pleasure. This can make reading an altogether more joyous experience, according to [45].

The first research question (RQ1) was what the UX goals should be for a social robot intended to motive children to read. The present study, due to its exploratory and small-scale nature, did not find any conclusive evidence that the book talks spurred intrinsic motivation to read. However, the identified UX goals were useful for creating an enjoyable interaction for most students. Student 1C's comment about the robot's incoherence causing unease initially, supports the idea that the robot must be effective enough at communicating verbally and non-verbally to entertain adaptive conversations, in order not to reduce the autonomy of the students [37], and to be able to guide and support them [14]. Hence, UX goal 1 is critical for enabling students in fulfilling their needs of competence and relatedness. Moreover, UX goals 2 and 3 emphasizing the role of the robot as a trustworthy peer creating a sense of relatedness were pivotal to give the students opportunities to show their competence and share their interpretations of the texts they had read. The idea of the robot being non-judgemental, which was emphasised in the expert review, is vital in this, as it encourages the students to share their personal thoughts and interpretations. The same can be said about the learning through teaching approach, as it allowed students to show their competence by teaching the robot. As illustrated by Freja in active mode, group 4 was clearly given the opportunity to give a nuanced explanation to the meaning of betraying someone's trust. It is difficult to assess the effect of the fourth UX goal. Furhat's ability to change appearance was

utilised to some small degree, but obviously not to its full potential, as the books chosen were all somewhat similar. The real effect of this ability remains to be tested in more detail.

The second research question (RQ2) was how a social robot should be designed to motivate children to read. Giving the impression of understanding emotions and having a personality can make the robot appear trustworthy [17, 33, 46]. Related to UX goal 4, there is no conclusive evidence at this point that being able to change characters is important for fulfilling the purpose of motivating children to read. However, the students clearly were curious about the characters. Actively engaging in the book can potentially contribute to autonomy and relatedness, which in turn contributes to motivation. The active mode was beneficial since it guided the children through the booktalk with more focus. Being able to talk about any book that the children choose to read supports autonomy and hence motivation. Designing the robot as a peer that the students can teach can likely facilitate relatedness, since they could take turns in teaching it about things it did not understand.

The third research question was whether the proposed design met the identified UX goals. The first UX goal was that the agent should be able to maintain adaptive conversations. We used a Wizard of Oz prototyping technique, and this makes it easier to reach this goal than it would be in an autonomously functioning implementated prototype. How well this objective can be achieved is yet an open empirical question. The second UX goal was that the agent should be a trustworthy peer. This goal was achieved in active mode where we could observe the more nuanced explanations. The third UX goal was to create a sense of relatedness among students, which is something the design achieved. The students talked to each other about the literature and about the robot characters. The fourth UX goal was to make the design immersive in the books, by means of characters. The empirical data does not give any indication that the character design contributed to immersion in the worlds of the books. What instead contributed to immersion was the active mode.

As many groups wanted to read “Black Belt”, of which there were too few copies, the prerequisite that the students could fulfil their need for autonomy by choosing a book of their own, could not be met. Nevertheless, the book talks showed potential. The students for the most part were happy to converse with the robot, which is an indication of trustworthiness. It also appeared to create a sense of relatedness.

In sum, this study suggests that robot book talks have the potential to bring out competence in, and create relatedness among, (Swedish) middle school students. If the students have the option to choose from a selection of books to read in groups, and discuss them with a social robot, they should also be able to fulfil their needs for autonomy. If the students need to be challenged, their tutors can always select new books for them to choose from.

The experience of interacting with a robot is unique for every individual. Therefore, it is crucial to design robots with the user experience in mind [7]. The book talks show that a robot like Furhat can participate in a classroom setting, while the students still focus on their task. Although the effect that such a social robot may have on the students’ well-being and reading ability over time has not been determined [47], there are reasons to believe that the book talks may create long-term well-being among students, because it seems to be able to contribute to satisfying the three basic psychological needs of autonomy, competence, and relatedness postulated by SDT. If a robot succeeds in doing so, the novelty effect may not be something to worry about [48].

5.2 Limitations

Because the study took place in a classroom, its ecological validity ought to be higher than its external validity. The results found in the study cannot be generalised to a larger population. To truly test the effect of a social robot in the book talk setting, it would have been necessary to include a control group, whose book talks would not have included the robot. Another option would have been to let the students have book talks without a robot as repeated measures design. Other limitations include that some participants saw through the wizard of oz set up, some students could not choose what book to read, and the characterisation of the agents was not very detailed. All three of these can pose a threat to internal validity of the study. The study was also negatively affected by some students (group 6) being absent from class during the first of the book talk sessions, thus preventing them from fully participating in the study. Because of the potential threat of the novelty effect against the ecological validity of this study, these initial results are not as trustworthy as long-term results. To truly test the qualities of the book talks, more sessions and more participants will be required. A long-term study could provide students with more structure, as explained by Vansteenkiste et al. [14]. In the current study, there was little room for guidance and constructive feedback. Finally, a potential threat to the study’s internal validity is social desirability bias, which describes a tendency to give answers to questions in ways that are not truthful, but rather make others perceive them as socially favourable [49]. This bias is hard to recognise without control evaluations such as the Children’s Social Desirability Short scale (CSD-S) [50], which might be useful in future studies.

5.3 Future Research

More long-term studies are required to investigate how much of the potential benefits of reading with robots depends on the novelty effect. Conducting longitudinal studies requires working prototypes or finished products that are ready for field testing. The effort and cost of develop such prototypes is large, and the present RtD study offers insights into considerations for the design of them. Storytelling that encourages students to use their imagination could contribute to competence [51, 52]. This would be particularly interesting in second language learning. We furthermore agree with Nicholson et al. [53], who argue that teachers should play a large part in the development process of educational technology. Co-design research with students and teachers in real school settings creates a unique overlap between education, human-robot interaction, and design.

5.4 Conclusion

The purpose of this study was to explore if and how talking to a robot about books might motivate fourth-grade students (10–11 years old) to read. While this study does not provide conclusive evidence that the designed robot book talks spurred intrinsic motivation, there are clear indications that book talks with an educational social robot can indeed provide opportunities for students to fulfill needs for competence and relatedness, which are essential for well-being and motivation. Provided that students can choose a book of their liking, and thus also fulfil their need for autonomy as well, robot book talks can be considered a promising approach to motivating fourth-grade students to read.

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Authors' Contributions MJ: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing - Original Draft. MA: Conceptualization, Writing - Review & Editing, Visualization, Supervision, Funding acquisition. ST: Supervision, Writing - Review & Editing. TZ: Writing - Review & Editing, Resources.

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Data Availability The datasets presented in this article are not readily available because of anonymity of the participants. Requests to access the datasets should be directed to registrator@liu.se.

Code Availability Not applicable.

Declarations

Ethics Approval Ethical review and approval was not required for the study on human participants in accordance with the local legislation and institutional requirements, since the study did not involve any physical intervention, it was not carried out with a method that aimed to affect the research participants physically or mentally, it did not involve any obvious risks, and it did not involve processing of sensitive personal data.

Consent to Participate The participants provided their informed consent to participate in this study verbally or over email. Children and their legal guardians provided written consent.

Consent for Publication The participants provided their informed consent to publish data from this study verbally or over email. Children and their legal guardians provided written consent.

Competing Interests The authors have no competing interests to declare that are relevant to the content of this article. This research involves human participants, but not animals. Informed consent was given by all participants and when relevant their legal guardians. The work in this article was based on the first author's master thesis. The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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